FRE-GT-9743 Assignment 6

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EUR/USD FX Forward Collateralised in GBP

Setup and notation

I take EUR/USD to mean USD per EUR. At maturity T, the long-EUR forward receives N euros and pays NK dollars, where K is the contractual strike. Let S(t) be the current USD price of one euro. I report the value in USD.

Write P(A|C) for the time-t value, in currency A, of one unit of A paid at T under collateral currency C. The common arguments (t,T) are suppressed. These are effective collateral-adjusted discount factors, not just the local OIS discount factors.

I assume full cash collateralisation, remuneration at the applicable overnight rate, a fixed GBP collateral currency, and no default, transaction costs or collateral-switch option. For the curve reconstruction in Q3, I explicitly use the deterministic funding-spread / FX-forward-invariance approximation.

1. Valuation formula

First value the two contractual payments using the same GBP collateral agreement. Receiving N EUR is worth N P(EUR|GBP) euros today; converting this present value at spot gives N S(t) P(EUR|GBP) dollars. The USD payment has present value NK P(USD|GBP). Therefore,

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V_USD(t) = N [S(t) P(EUR|GBP) - K P(USD|GBP)].
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The opposite trade has the negative of this value. The collateral currency determines the effective discount curves; it does not change the currency of the contractual payments. Using SONIA alone to discount both payments would omit their currency funding basis.

Equivalently, under the USD pricing measure, define c_USD as the USD collateral rate and b_USD,GBP as the additional spread needed for GBP collateral. The effective discounting rate is c_USD + b_USD,GBP, so [1]

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V_USD(t) = N E_t^Q_USD[exp(-integral_t^T
(c_USD(u) + b_USD,GBP(u)) du) * (S(T) - K)].
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This expectation is the general pricing statement. The two effective bond prices in the first formula already contain the relevant funding and dependence effects.

2. Fair forward strike

The fair strike makes the initial value zero. Since P(USD|GBP) is positive,

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K_fair = S(t) P(EUR|GBP) / P(USD|GBP).
V_USD(t) = N P(USD|GBP) (K_fair - K).
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Thus a long-EUR forward has positive value when its contractual strike is below the current fair forward. At maturity the discount factors become one and the value reduces to N(S(T)-K), as required.

3. Are the supplied discount factors sufficient?

In the usual deterministic curve framework, yes, provided the current EUR/USD spot rate and the contract terms are also known. I interpret df[GBP,USD] as P(GBP|USD) and df[EUR,USD] as P(EUR|USD). The first currency is the payment currency and the second is the collateral currency.

For compactness, define

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g = df_SONIA = P(GBP|GBP),
u = df_SOFR = P(USD|USD),
e = df[EUR,USD] = P(EUR|USD),
h = df[GBP,USD] = P(GBP|USD).
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FX-forward invariance means that changing the collateral currency leaves the fair FX forward unchanged within this curve approximation [2]. Applying it first to GBP/USD gives

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P(GBP|GBP) / P(USD|GBP) = P(GBP|USD) / P(USD|USD).
P(USD|GBP) = g u / h.
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Next apply it to EUR/USD:

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P(EUR|GBP) / P(USD|GBP) = P(EUR|USD) / P(USD|USD).
P(EUR|GBP) = g e / h.
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Substituting these results into Q1 and Q2 gives

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V_USD(t) = N (g / h) [S(t) e - K u].
K_fair = S(t) e / u.
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The GBP collateral adjustment g/h scales the off-market value of this trade, but cancels from its fair strike in this approximation. The EuroSTR discount factor is not additionally needed once the effective EUR-under-USD curve e has already been built. It can still be an input to that curve's construction.

As a consistency check, at K = S(t)e/u the value is zero. If cross-currency funding basis is zero, h = g and e is the EUR OIS discount factor. The formulas then reduce to the ordinary covered-interest-parity result.

The qualification matters: FX-forward invariance across different collateral currencies is an assumption, not a general consequence of triangular FX arbitrage. In a model with stochastic currency funding spreads and dependence on FX and interest rates, the five initial discount curves alone need not identify P(EUR|GBP) and P(USD|GBP). In that setting I would need those GBP-collateral curves directly, or a calibrated joint model / additional market information. The general formulas in Q1 and Q2 still apply [1].

My conclusion is therefore: the supplied curves are sufficient for the standard deterministic construction, with spot FX supplied separately; they are not unconditionally sufficient for an unrestricted stochastic collateral model.

References

[1] Fujii, M. and Takahashi, A. (2015). A General Framework for the Benchmark Pricing in a Fully Collateralized Market, Sections 2 and 5.1. https://arxiv.org/abs/1508.06339

[2] Burgess, N. Synthetic Forward FX Replication & Collateralized Discount Factors, slides 6 and 9-10. https://nicholasburgess.co.uk/wp-content/uploads/2025/04/synthetic-fx-forward-replication.pdf